Tutorial Kontrol LED ESP32 (GPIO 19 & 21) dengan PictoBlox

🎯 Tujuan

Siswa dapat:

- Mengontrol 2 LED secara ON/OFF
- Menggunakan PictoBlox untuk mengirim perintah ke ESP32

🧰 Alat & Bahan

- ESP32 (disarankan DOIT / DevKit)
- 2 LED
- 2 Resistor (220Ω / 330Ω)
- Breadboard
- Kabel jumper
- Laptop + PictoBlox

🔌 Wiring (Rangkaian)

Hubungkan seperti ini:

LED 1

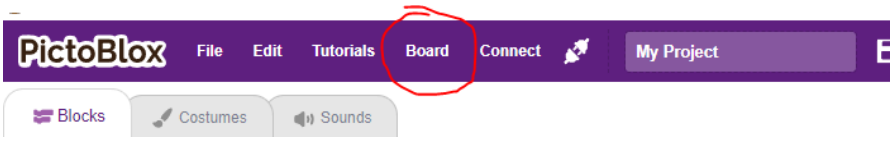
- Kaki panjang LED → GPIO **19**
- Kaki pendek → resistor → GND

LED 2

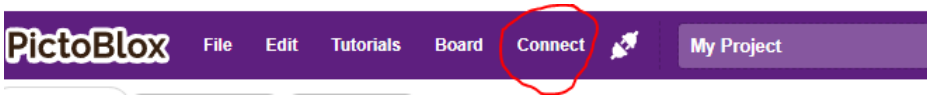
- Kaki panjang LED → GPIO **21**
- Kaki pendek → resistor → GND

💻 Setup PictoBlox

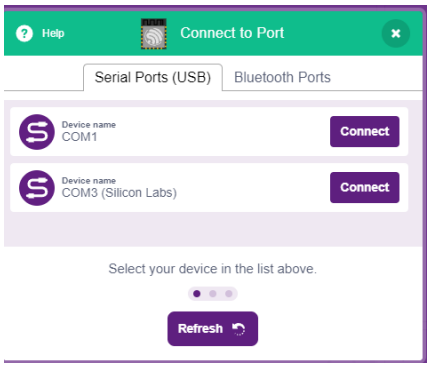
1. Buka PictoBlox
2. Pilih **Board** → **ESP32**



3. Klik **Connect**



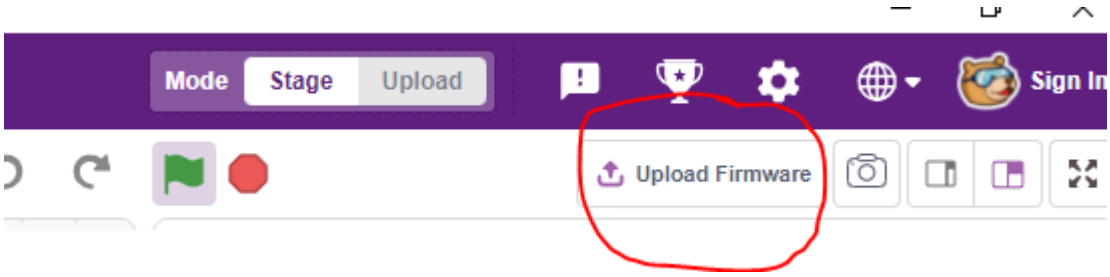
4. Pilih port ESP32



5. Mode:

gunakan **Upload Mode** atau **Realtime Mode** (disarankan Realtime untuk belajar)

6. Upload Firmware untuk pertama kali melakukan program

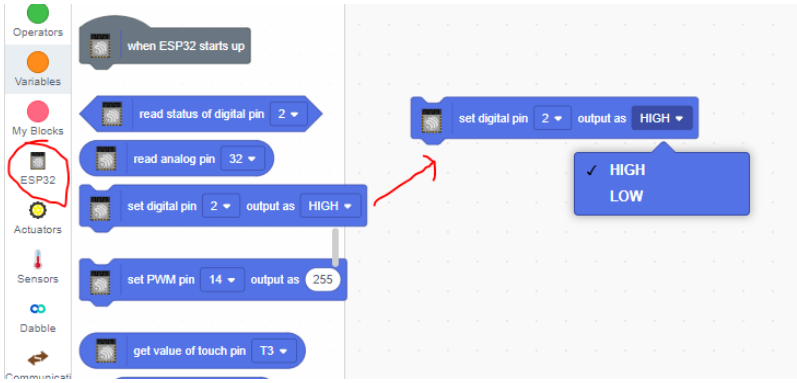


🧠 Logika Program

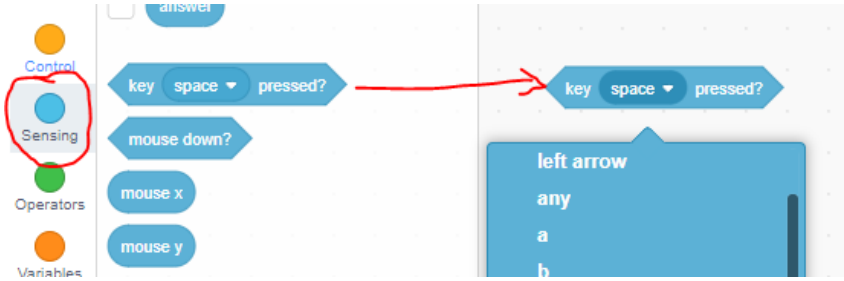
Kita akan:

- Mengirim perintah dari PictoBlox (misalnya tombol keyboard)
- ESP32 membaca perintah
- LED menyala atau mati

Block yang digunakan untuk digital ouput dan cek keyboard:



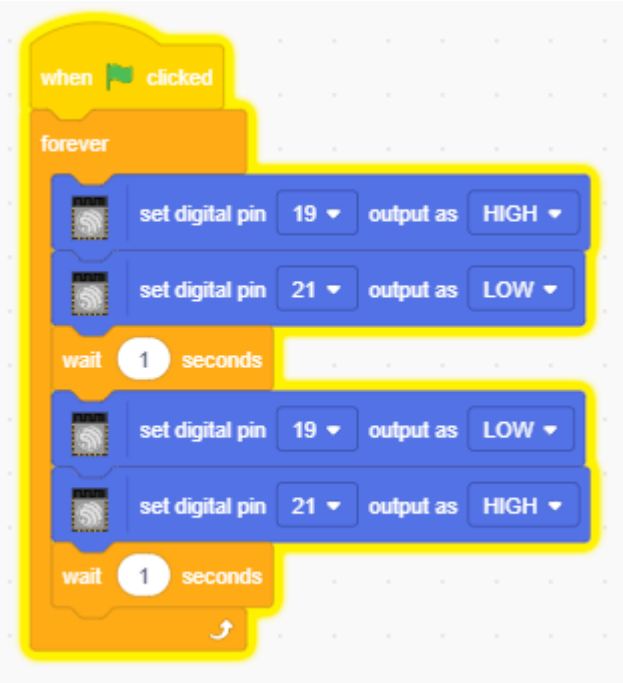
Set digital pin adalah nomor GPIO LED, HIGH untuk menyalakan, LOW untuk mematikan LED



Block yang digunakan untuk mengetahui keyboard yang ditekan

Block Program (Dasar) Control Led Secara Bergantian

Led_1 = hidup, Led_2 = mati selama 1 detik dan Led_1 = mati, Led_2 = hidup selama 1 detik, terus menerus.

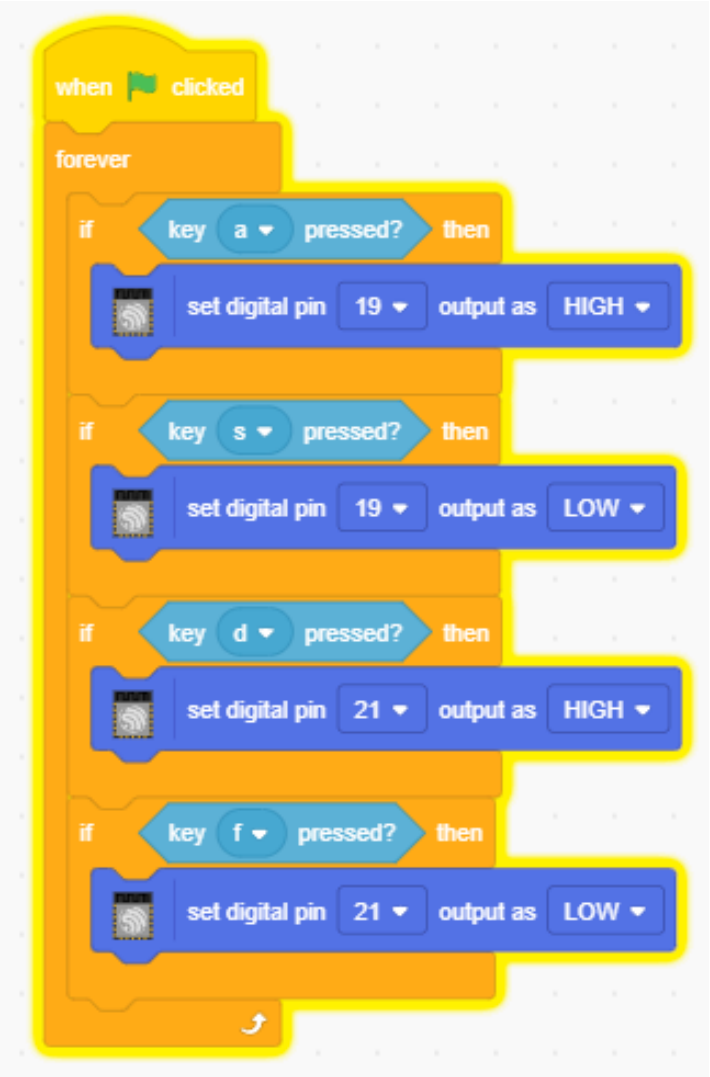


Blok Program (Lanjutan)

🔵 Saat tombol ditekan

LED 1 (GPIO 19)

- tekan tombol **A** → **LED ON**
- tekan tombol **S** → **LED OFF**

LED 2 (GPIO 21)

- tekan tombol **D** → **LED ON**
- tekan tombol **F** → **LED OFF**